Factor Pricing Models

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Review

CAPM:

$$R_A - R_f = \beta_A \times \left(R_M - R_f\right)$$

- Is this a compelling economic story?
- Why does Apple's price go up?
- Is it a good pricing model? Would you use it to trade with?
  - $\beta_A \neq 0$: Market is priced risk factor
  - $r^2 \approx 20\%$: Market is not the only risk factor

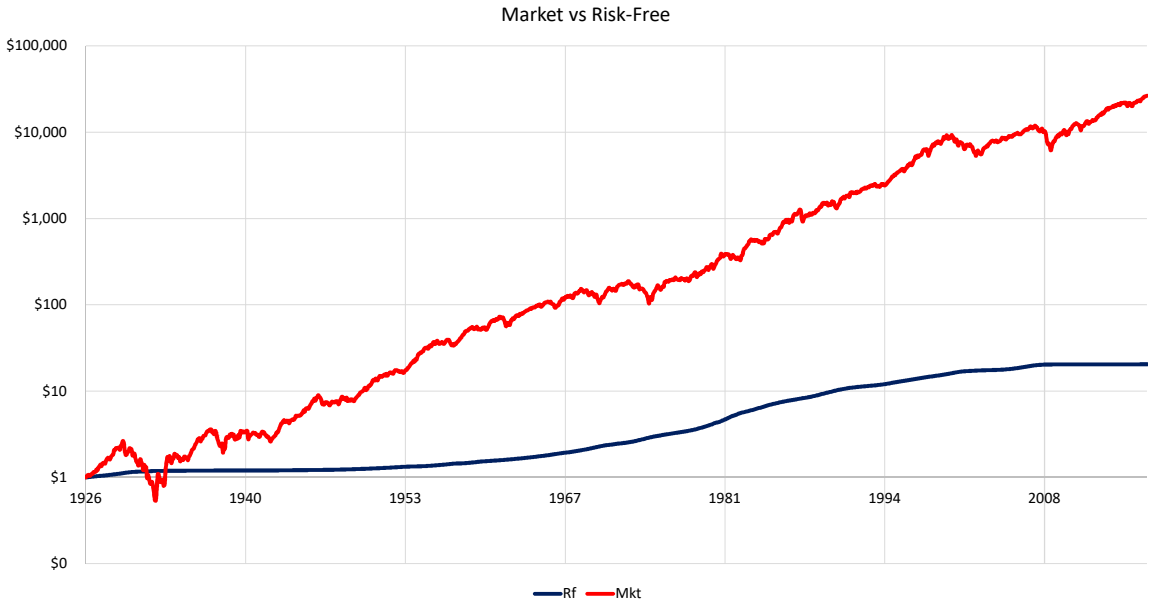
Overview

- CAPM is an economic theory
  - We tested it empirically using regression analysis
  - Economic model → statistical model

- As economists we want a compelling economic model
  - What are the true, fundamental, economic sources of risk?
  - GDP, wages, income, productivity?

- Use statistics to identify what risk factors work well
  - Market, size, value
  - Statistical model → what does it imply economically?

Current Event: Risk Factors



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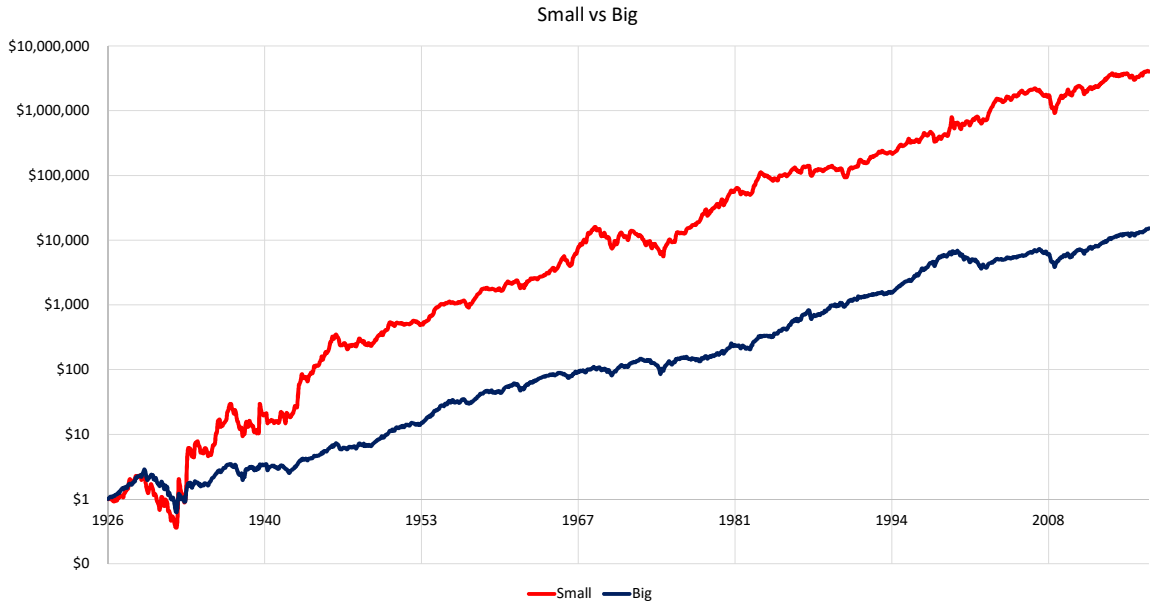


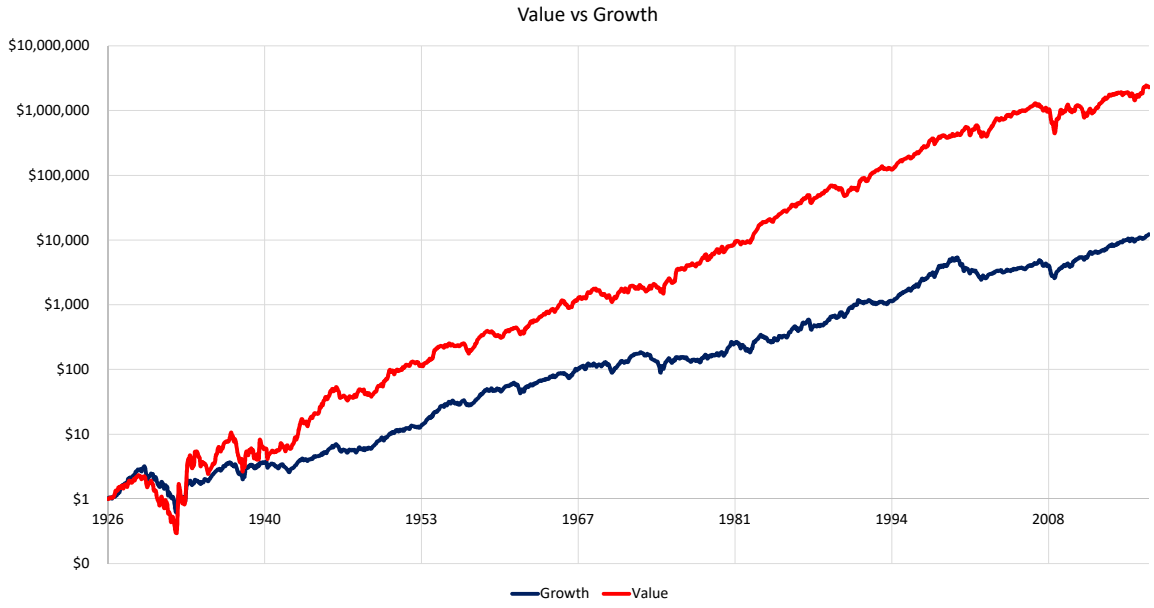
Market vs Book Value of equity

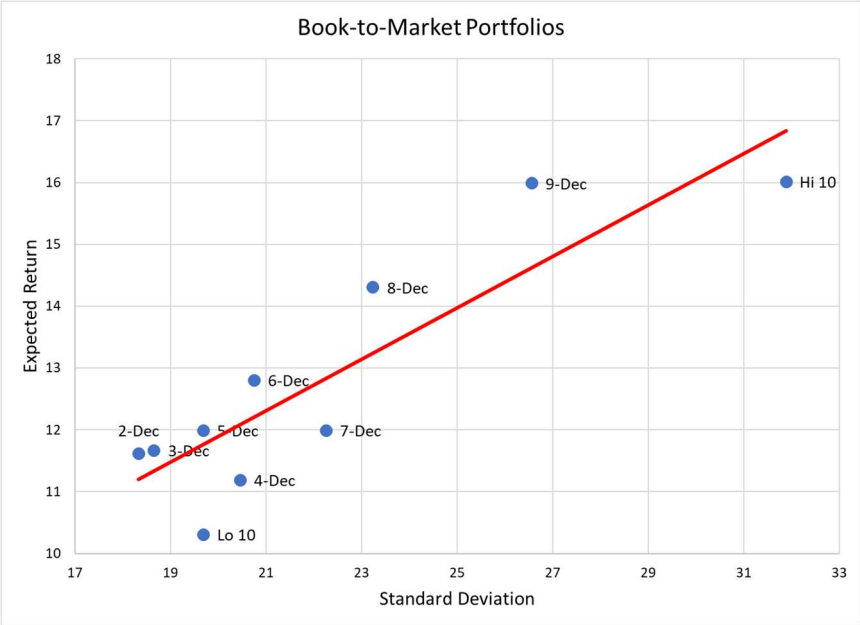
	Optimistic	Pessimistic
$\frac{B}{M}$ **High: Value**		
$\frac{B}{M}$ **Low: Growth**		

Market vs Book Value of equity

	Optimistic	**Pessimistic**
$\frac{\mathbf{B}}{\mathbf{M}}$ **High: Value**	Good deal	Bad news
$\frac{\mathbf{B}}{\mathbf{M}}$ **Low: Growth**	Good news	Overpriced

Current Event: Risk Factors





Value Premium Anomaly

- Value stocks outperform growth stocks on average
  - ⇒ Value stocks are riskier than growth stocks
  - What is the risk premium investors are earning for holding value over growth?
  - Value premium anomaly

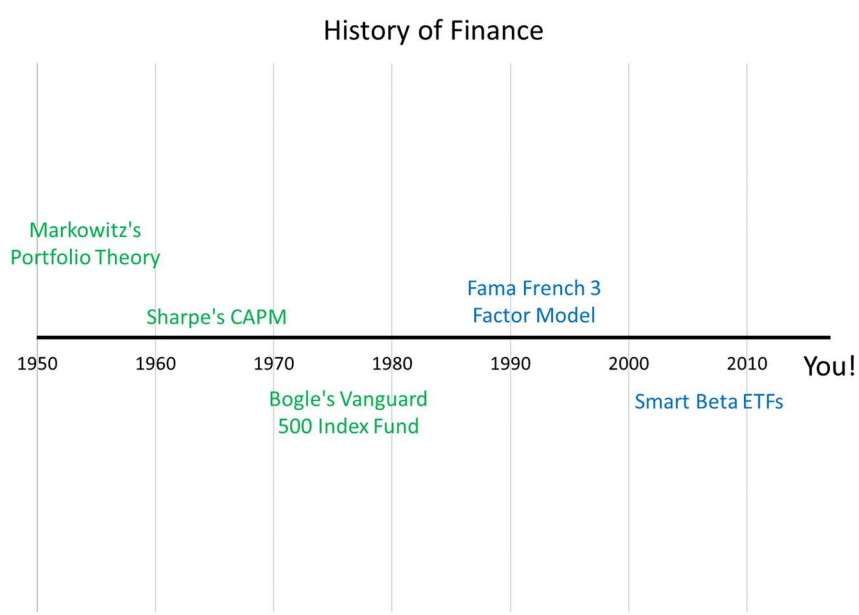
	Optimistic	**Pessimistic**
$\frac{\mathbf{B}}{\mathbf{M}}$ **High: Value**	Good deal	~~Bad news~~
$\frac{\mathbf{B}}{\mathbf{M}}$ **Low: Growth**	~~Good news~~	Overpriced

Discovering Factors

To find new factors, we can use F-tests to compare asset pricing models

- Example: FF3 vs. CAPM

$$Full: R_i = \alpha + \beta_M \cdot R_M + \beta_{Size} \cdot R_{Size} + \beta_{Value} \cdot R_{Value}$$
$$Reduce: R_i = \alpha + \beta_M \cdot R_M$$

- Hypothesis
  - Null: $\beta_{Size} = \beta_{Value} = 0$
  - Alternative: At least one of β_{Size} or β_{Value} are not 0

- Economic conclusion
  - Null ⇒ CAPM is the better asset pricing model (i.e. Size and Value factors are not priced)
  - Alternative ⇒ FF3 is the better asset pricing model (Size and Value factors are priced)